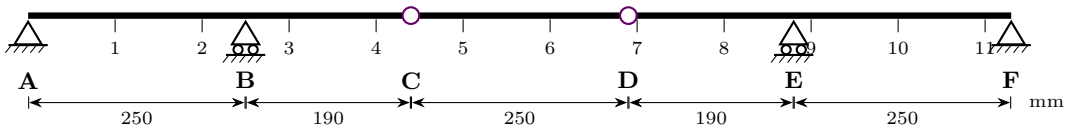
Predict – scan – check

Lab S2 Influence Lines: in-class activity (15 minutes, in pairs)

Name(s): _____ Group: _____

Write your prediction *first*, in ink. Then scan the code (it opens the *Explorer* tab of the Lab S2 course page), test it, and write what you found and why. A wrong prediction with a good explanation of what you learned earns full credit.

Lab S2 page, Explorer tab



1. **Sign of R_A.** (Explorer: press *Unit load* under *Loads*.) A single mass stands at halt 4 (on the cantilever, 150 mm past B). Is the reaction at A upward, zero or downward? Prediction and reason:

   Explorer result (η_A at $x = 400$): _____ A negative reaction means: _____

2. **Largest R_B.** Where must a single mass stand for the largest reaction at B, and will R_B be smaller or larger than the mass itself? Prediction:

   Explorer: position _____, η_B = _____. Why can a reaction exceed the load? _____

3. **The far side.** The mass is at halt 8 (between D and E). Predict R_A and R_B, then check. What does the hinge at D transmit when the suspended span is unloaded?

4. **Compound load.** Press *Test 2 pair* under *Loads* (5 kg and 2.5 kg, 100 mm apart). Predict the position of the 5 kg mass that gives the most negative R_A and the value in kgf; then find it by trial.

   Explorer: position _____, R_A = _____ kgf. Which mass stands on the vertex C, and why does that give the extreme? _____

5. **Distributed load.** Tick *shade the areas* for R_A. Predict the sign of R_A when a uniform load covers the whole model; then read the three areas and the net value.

   Explorer: areas +_____ mm and _____ mm, net _____ mm, so R_A = _____ w.

In the laboratory: after Test 1, open the *Lab* tab of the same page and type your gauge-A readings into the Test 1 table. Note the largest absolute difference from theory and at which halt it occurs: _____ at halt _____. Is it within the ±0.03 expected from a ±5 mm placement error and ±1 division? _____